Time-independent Schrodinger equation

Consider the Schrodinger equation

$$i\hbar\frac{\partial}{\partial t}\psi(x,t) = H\psi(x,t)$$

Let $\psi(x,t)$ be separable as

$$\psi(x,t) = \varphi(t)\phi(x)$$

Let it be the case that for the time derivative we have

$$i\hbar\frac{d}{dt}\varphi(t) = E\varphi(t)$$

where E is a constant. Then

$$i\hbar\frac{\partial}{\partial t}\psi(x,t) = E\varphi(t)\phi(x) = E\psi(x,t)$$

Hence by the Schrodinger equation

$$E\psi(x,t) = H\psi(x,t)$$

If H is independent of t then $\varphi(t)$ cancels and we have the time-independent Schrodinger equation

$$E\phi(x) = H\phi(x)$$

Summarizing, let $\psi(x,t)$ be a wavefunction such that

1. $\psi(x,t)$ solves the Schrodinger equation
2. $\psi(x,t)$ is separable as $\varphi(t)\phi(x)$
3. The time derivative of $\varphi(t)$ satisfies $i\hbar\, d\varphi(t)/dt = E\varphi(t)$ where E is a constant
4. H is independent of t

Then $\phi(x)$ solves the time-independent Schrodinger equation

$$E\phi(x) = H\phi(x)$$

For example, the Hamiltonian for the harmonic oscillator is

$$H = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \frac{1}{2}m\omega^2 x^2$$

The wavefunctions are

$$\psi_n(x,t) = C_n \exp\left(-\frac{iE_n t}{\hbar}\right)\exp\left(\frac{m\omega x^2}{2\hbar}\right)\frac{d^n}{dx^n}\exp\left(-\frac{m\omega x^2}{\hbar}\right)$$

where C_n is a normalization constant and

$$E_n = \hbar\omega\left(n + \tfrac{1}{2}\right)$$

Taking the time derivative of $\psi_n(x,t)$ we find

$$i\hbar\frac{\partial}{\partial t}\psi_n(x,t) = E_n\psi_n(x,t)$$

Hence

$$E_n\psi_n(x,t) = H_n\psi_n(x,t)$$

Since H is independent of t we have

$$E_n\phi_n(x) = H\phi_n(x)$$

where

$$\phi_n(x) = C_n \exp\left(\frac{m\omega x^2}{2\hbar}\right)\frac{d^n}{dx^n}\exp\left(-\frac{m\omega x^2}{\hbar}\right)$$

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